King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 253

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Let $z = \tan^{-1}\left(\frac{x}{y}\right)$, $x = te^s$, $y = s \ln t$. The value of $\frac{\partial z}{\partial t}$ when $(s,t) = (1,e)$ is equal to

(a) $\frac{e-1}{e^4+1}$

(b) $\frac{2e}{e^4+1}$

(c) 0

(d) $\frac{-e}{e^4+1}$

(e) $\frac{e+1}{e^4+1}$

2. If (a,b,c) is the point of intersection of the plane $x+2y-z=-9$ and the line through $(2,-1,3)$ that is perpendicular to this plane, then $abc =$

(a) 0

(b) -9

(c) 6

(d) -12

(e) -6

3. The range of the function $f(x,y) = 2e^{xy}$ is

(a) $(3,\infty)$

(b) $[2,\infty)$

(c) $(-\infty,\infty)$

(d) $(0,\infty)$

(e) $[1,\infty)$

4. If $y = Ax + B$ is the level curve of the function $f(x, y) = \ln(x - y)$ that passes through $(e, 0)$, then $AB =$

(a) 0

(b) e

(c) $-e$

(d) $3e$

(e) $-3e$

5. If $f(x, y, z) = y^2 e^{xz^2}$, then $\dfrac{\partial^3 f}{\partial z\, \partial y\, \partial x} =$

(a) $2yz(xz^2 + 1)e^{xz^2}$

(b) $2xz(z^4 + 1)e^{xz^2}$

(c) $4yz(xz^2 + 1)e^{xz^2}$

(d) $xyz(z^2 + 1)e^{xz^2}$

(e) $4xy(z^3 + 1)e^{xz^2}$

6. If $f(x, y) = \ln\left(\dfrac{x + y}{x - y}\right)$, then $xf_y(x, y) + yf_x(x, y) =$

(a) $-\dfrac{1}{2}$

(b) 0

(c) -2

(d) 1

(e) 2

7. The set of all points where $f(x, y, z) = \dfrac{\sin z}{e^x - e^y}$ is continuous is given by

(a) $\{(x, y, z) : x, y, z \in (-\infty, \infty)\}$

(b) $\{(x, y, z) : x = y\}$

(c) $\{(x, y, z) : x \neq y,\ z \in (-\infty, \infty)\}$

(d) $\{(x, y, z) : x, y \in (-\infty, \infty),\ z \in [-1, 1)\}$

(e) $\{(x, y, z) : x \neq y,\ z \in [-1, 1]\}$

8. If $x + by + cz = d$ is an equation of the plane that contains the lines

$$\frac{x-1}{-2} = y - 4 = z \quad \text{and} \quad \frac{x-2}{-3} = \frac{y-1}{4} = \frac{z-2}{-1},$$

then $b + c + d =$

(a) 5

(b) 0

(c) -4

(d) 7

(e) -3

9. Which one of the following statements is FALSE about the graph of the equation $4x^2 - 4y + z^2 = 0$?

(a) Its trace in the plane $y = 1$ is an ellipse

(b) Its trace in the xz-plane is the origin

(c) Its trace in the plane $y = -1$ is an ellipse

(d) It is an elliptic paraboloid

(e) Its trace in the xy-plane is a parabola

10. $\lim_{(x,y)\to(0,0)} \left[\frac{1}{\ln(1+x^2+y^2)} - \frac{1}{x^2+y^2}\right] =$

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) 2

(e) The limit does not exist

11. The distance between the parallel planes

$$x - 3y + 2z = 1$$

$$2x - 6y + 4z = 3$$

is equal to

(a) $\dfrac{\sqrt{14}}{28}$

(b) $\dfrac{\sqrt{14}}{7}$

(c) $\dfrac{\sqrt{14}}{2}$

(d) $\dfrac{\sqrt{14}}{4}$

(e) $\dfrac{\sqrt{14}}{14}$

12. Let $z = xy + \dfrac{y}{x}$. Using differentials, the change in z when (x, y) moves from the point $(2, 1)$ to the point $(2.1, 1.05)$ is approximately equal to

(a) $\dfrac{1}{40}$

(b) $\dfrac{1}{5}$

(c) $\dfrac{2}{5}$

(d) $\dfrac{3}{40}$

(e) $\dfrac{3}{5}$

13. The graph of the equation

$$x^2 + 2y^2 - z^2 + 2x + 6z = k^2 + k + 8$$

is

(a) an elliptic paraboloid if $k = 0$ and $k = -1$

(b) a hyperboloid of one sheet if $-1 < k < 0$

(c) a hyperboloid of one sheet for $k > -1$

(d) an elliptic cone if $k > 0$

(e) a hyperboloid of two sheets if $-1 < k < 0$

14. If (a, b, c) is the point of intersection between the lines

$$L_1 : x = 2 + 4t,\ y = 3,\ z = 1 - t$$

$$L_2 : x = 2 + 2s,\ y = 3 + 2s,\ z = 1 + s$$

then $abc =$

(a) -4

(b) -10

(c) 8

(d) 6

(e) 0

15. If the equation $z^3 - 2xy + 2yz + y^3 - 2 = 0$ defines z implicitly as a differentiable function of x and y, then the value of $\dfrac{\partial z}{\partial y}$ when $(x, y, z) = (1, 1, 1)$ is equal to

(a) $\dfrac{4}{5}$

(b) $-\dfrac{2}{5}$

(c) $-\dfrac{3}{5}$

(d) $\dfrac{2}{5}$

(e) 0

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$z_t = \frac{e}{1+e^4} - \frac{e}{1+e^4} = 0$.
2	**(d)**	$t = -1$ gives $(1, -3, 4)$, so $abc = -12$.
3	**(d)**	xy is any real number, so $2e^{xy}$ takes all values in $(0, \infty)$.
4	**(c)**	$\ln(x - y) = 1 \Rightarrow y = x - e$, so $AB = -e$.
5	**(c)**	$f_x = y^2 z^2 e^{xz^2}$, $f_{xy} = 2yz^2 e^{xz^2}$, $f_{xyz} = 4yz(xz^2 + 1)e^{xz^2}$.
6	**(e)**	$f_x = \frac{-2y}{x^2-y^2}$, $f_y = \frac{2x}{x^2-y^2}$, so $xf_y + yf_x = 2$.
7	**(c)**	Denominator zero iff $x = y$; $\sin z$ is fine for all z.
8	**(d)**	$\mathbf{n} \parallel \langle 1, 1, 1 \rangle$, point $(1, 4, 0)$: $x + y + z = 5$, so $b + c + d = 7$.
9	**(c)**	At $y = -1$: $4x^2 + z^2 = -4$ has no points, so it is not an ellipse.
10	**(b)**	With $w = x^2 + y^2$ and two uses of L'Hopital the limit is $\frac{1}{2}$.
11	**(a)**	Point $(1, 0, 0)$: $\dfrac{\lvert 2 - 3 \rvert}{\sqrt{56}} = \dfrac{\sqrt{14}}{28}$.
12	**(b)**	$dz = \frac{3}{4}(0.1) + \frac{5}{2}(0.05) = \frac{1}{5}$.
13	**(e)**	$(x + 1)^2 + 2y^2 - (z - 3)^2 = k(k + 1) < 0$ for $-1 < k < 0$: two sheets.
14	**(d)**	$s = t = 0$ gives the point $(2, 3, 1)$, so $abc = 6$.
15	**(c)**	$z_y = -F_y/F_z = -\dfrac{3}{5}$ at $(1, 1, 1)$.